

Ethan Deal

+1 (202) 560-2873 • ecd31@duke.edu • [linkedin.com/in/ethdeal](https://www.linkedin.com/in/ethdeal) • ethdeal.github.io

Full-Stack Software Engineer | React, TypeScript, Python, FastAPI, Node.js, RAG Systems

EDUCATION

Duke University & Duke Kunshan University | Dual Degree
B.S. Computation and Design, Computer Science

May 2026 | Kunshan 昆山, CN

- GPA: 3.6/4.0; Awarded Merit Scholarship (60% Tuition)

EXPERIENCE

BRIYA | Fullstack Developer

July - Aug 2025 | Washington, DC

Previous role: IT Specialist

May - Aug 2024

- Built and deployed a **MERN** stack room booking system supporting **5 campuses** and **100+ faculty**, achieving **zero downtime** since launch.
- Developed Node.js/MongoDB REST APIs for booking CRUD operations, recurring events, and conflict detection.
- Designed a React interface for desktop and mobile with multiple calendar views, drag-and-drop editing, and intuitive booking workflows.
- Deployed frontend and backend with GitHub Pages, Linode, SSL, Nginx reverse proxying, PM2, and DNS configuration.
- Previously enrolled and configured 1000+ Chromebooks, performed disk wipes, and supported device troubleshooting.

DUKE UNIVERSITY | Data Automation Engineer

May - July 2025 | Durham, NC

- Built a Python/Pandas batch processing script to clean and segment **10,000+ mailing records**, exporting ready-to-use CSV reports.
- Developed a Python ETL tool to consolidate multi-event datasets into venue-based Excel reports for cross-event analysis.
- Automated Qualtrics workflows to centralize large-scale survey responses into standardized spreadsheets.

VINTORY | Generative AI Design Specialist

May - Sep 2023 | Remote

- Built a Midjourney prompt library and ChatGPT workflow to generate structured prompt arrays for branded design production.
- Created **30+** postcards and **100+** templates/landing pages for vacation rental management clients, using AI-assisted design workflows.

PROJECTS

Ember Intelligence

Feb 2026 - Present

- Built and deployed a multi-tenant B2B knowledge retrieval platform that connects enterprise data sources to AI agents across finance and consulting teams. Piloted across multiple organizations, supporting **50+ users**.
- Developed MCP server-driven integrations connecting AI harnesses to sources including Google Drive, SharePoint, Salesforce, and Harvest.
- Engineered a provider-agnostic Python ingestion pipeline with OAuth source connections, document chunk enrichment, ACL-aware metadata, and Qdrant indexing for hybrid retrieval.
- Designed a stateful incremental sync architecture for tracking changes, deleted content, retries, reconciliation, and selective re-indexing.
- Built fullstack features across Next.js, FastAPI, PostgreSQL/Supabase, Nango, Cohere, and Qdrant for source management, retrieval, and chat workflows.

RAG Fact-Checking Extension

July 2025 - March 2026

- Designed a Chrome extension in React with a Node.js/Express backend to analyze highlighted claims in under **5 seconds**.
- Architected a **RAG** pipeline using **LangChain**, ChromaDB, and a locally hosted MiniLM embedding model.
- Benchmarked single-step agentic vs. deterministic two-step retrieval architectures, evaluating latency, retrieval accuracy, context leakage, and susceptibility to prompt-injected misinformation.

FlightOps Database Design

Oct 2024 – Jan 2025

- Designed a MySQL airline operations database with 10+ interrelated tables, SQL views, stored procedures, and CRUD workflows for booking, routing, assignment, and maintenance queries.
- Built a Playwright scraping pipeline to query live flight records and compiled datasets of **10,000+** airports, **7,000** aircraft, and **600** airlines.

SKILLS

Languages: Python, TypeScript, SQL, HTML/CSS, Java

Fullstack: React, Next.js | Node.js, Express, FastAPI | PostgreSQL, MongoDB, MySQL

AI & Retrieval: LangGraph, ChromaDB, Qdrant, Cohere, LiteLLM

Cloud & Tools: Docker, Git, Supabase, Google Cloud, Linode

ADDITIONAL INFORMATION

Languages: English (native), Mandarin (intermediate), Spanish (beginner)

Interests: Ultimate frisbee, digital painting, travel photography, piano, guitar, soccer